

Negative Autocorrelation in *Micro* MidPrice Differences

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High-frequency financial data, particularly the differences (returns) of the Midprice, consistently exhibit significant negative first-order autocorrelation. *Why?*

1 Differenced White Noise Correlation

Let $\{U_t\}$ be a pure white noise process with:

$$\mathbb{E}[U_t] = 0, \quad \text{Var}(U_t) = \sigma^2 > 0, \quad \text{Cov}(U_t, U_{t-k}) = 0 \quad \forall k \neq 0. \quad (1)$$

The first-order difference of the white noise process is:

$$Z_t = U_t - U_{t-1}. \quad (2)$$

We compute the covariance between Z_t and its first lag Z_{t-1} :

$$\begin{aligned} \text{Cov}(Z_t, Z_{t-1}) &= \text{Cov}(U_t - U_{t-1}, U_{t-1} - U_{t-2}) \\ &= -\text{Var}(U_{t-1}) = -\sigma^2. \end{aligned} \quad (3)$$

All cross-terms vanish due to the white noise properties. The variance of Z_t is:

$$\text{Var}(Z_t) = \text{Var}(U_t) + \text{Var}(U_{t-1}) = 2\sigma^2. \quad (4)$$

Therefore, the first-order autocorrelation coefficient is:

$$\rho_Z(1) = \frac{\text{Cov}(Z_t, Z_{t-1})}{\sqrt{\text{Var}(Z_t)\text{Var}(Z_{t-1})}} = \frac{-\sigma^2}{2\sigma^2} = -\frac{1}{2}. \quad (5)$$

Any white noise series, once differenced, possesses a fixed theoretical first-order autocorrelation of exactly $-1/2$. This is an inherent "saw-tooth" effect introduced by the differencing operator.

2 The Price Model with Microstructure Noise

We now embed the previous result into a standard financial price process, following the reduced-form framework commonly used in market microstructure literature [ASMZ05].

2.1 Observed Price vs. Efficient Price

Let X_t denote the unobservable "efficient" log-price, assumed to follow a pure random walk (drift is negligible at high frequencies):

$$dX_t = \sigma dW_t, \quad (6)$$

where σ is the volatility and W_t is standard Brownian motion. Let U_t represent the market microstructure noise (i.i.d., mean zero, variance a^2), arising from bid-ask spreads, price discreteness (tick size), and rounding errors. The observed log-price (e.g., the log of the Midprice) is:

$$\tilde{X}_t = X_t + U_t. \quad (7)$$

Let the sampling interval be Δ . The observed high-frequency return is:

$$Y_i = \tilde{X}_i - \tilde{X}_{i-1} = (X_i - X_{i-1}) + (U_i - U_{i-1}). \quad (8)$$

The true price increment has variance $\sigma^2\Delta$, while the noise component is the difference of two i.i.d. noise terms. Since the efficient returns are uncorrelated across time, the autocovariance of Y_i is solely determined by the overlapping noise term:

$$\begin{aligned} \text{Cov}(Y_i, Y_{i-1}) &= \text{Cov}(U_i - U_{i-1}, U_{i-1} - U_{i-2}) \\ &= -\text{Var}(U_{i-1}) = -a^2. \end{aligned} \quad (9)$$

2.2 Total Variance of Observed Returns

The variance of the observed returns is the sum of the efficient price variance and the noise variance:

$$\text{Var}(Y_i) = \sigma^2 \Delta + 2a^2. \quad (10)$$

Combining Eq. 9 and Eq. 10, the theoretical first-order autocorrelation of the observed returns is:

$$\rho_Y(1) = \frac{\text{Cov}(Y_i, Y_{i-1})}{\text{Var}(Y_i)} = \frac{-a^2}{\sigma^2 \Delta + 2a^2}. \quad (11)$$

Define the proportion of the total return variance attributable to microstructure noise as:

$$\pi(\Delta) = \frac{2a^2}{\sigma^2 \Delta + 2a^2}. \quad (12)$$

As we sample at increasingly higher frequencies (e.g., tick-by-tick or millisecond data), the sampling interval $\Delta \rightarrow 0$. Consequently, $\lim_{\Delta \rightarrow 0} \pi(\Delta) = 1$. This implies that in the ultra-high-frequency limit, nearly 100% of the observed return variance is driven by microstructure noise, completely swamping the true price signal. Substituting $\Delta \rightarrow 0$ into Eq. 11 yields the asymptotic lower bound for the autocorrelation: $\lim_{\Delta \rightarrow 0} \rho_Y(1) = -1/2$.

References

- [ASMZ05] Yacine Ait-Sahalia, Per A Mykland, and Lan Zhang. How often to sample a continuous-time process in the presence of market microstructure noise. *The review of financial studies*, 18(2):351–416, 2005.